

SARTHAK BICHHAWA

Senior Data Scientist · Product Analytics, Measurement & Applied AI

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SUMMARY

Data scientist with **9+ years** building product measurement, trusted reporting and applied-AI systems. Leads **cross-team metric design**, telemetry investigations and **validation controls** that turn complex data into executive decisions across cloud reliability, revenue attribution and customer health.

HIGHLIGHTS

- Led customer-health measurement across **8+ teams**; accelerated insights from **3-5 days to near real-time**.
- Recovered **~\$45M/month in partner revenue attribution**, with **validation gates** and independent Finance reconciliation.
- Co-developed data governance across **400+ production datasets**, with hourly profiling, quality gates and **LLM-powered lineage**.

EXPERIENCE

Microsoft — Azure Storage | Senior Data Scientist

Nov 2020 – Present

Previously Data Scientist. Own revenue-critical data platforms, customer-health analytics and applied-AI systems for Azure Storage.

- Designed **customer-health metrics** and drove executive reviews across **8+ engineering/support teams**, with **50% fewer** long-aging issues and **16% faster** mitigation. Accelerated detection and action on customer-impacting issues, helping retain **10+ at-risk accounts** worth millions per month each.
- Recovered **~\$45M/month in partner revenue attribution** after upstream data loss broke a pipeline attributing **\$5B+/year**. Traced a **20+ component** legacy pipeline, sourced alternate telemetry and drove **six months of backfills** through quarter close (7 incidents); led a **severity-2 incident across 4 teams**. Established validation gates, independent Finance reconciliation and multi-partner credit.
- Co-developed a **self-service governance portal** with freshness, volume and schema anomaly checks, quality gates and hourly profiling of **400+ production datasets**. Developed its **LLM lineage engine** to map upstream/downstream dependencies from pipeline code across sources, unifying lineage, ownership, usage and job health previously assembled through hours of manual investigation.
- Built a production customer-health LLM agent connecting **more than 15 live-telemetry signal types** to answer questions in plain English, cutting executive-review preparation from **days to under 2 minutes**.
- Shipped an Azure OpenAI **multi-agent incident investigator** with planner/executor orchestration, confidence scoring and run tracing, cutting diagnosis from **hours to minutes**; built **four reusable Model Context Protocol (MCP) servers** for governed telemetry access.
- Designed adaptive infrastructure stress tests using **multi-armed bandits** and outcome-tuned scoring across thousands of simulations per release; reduced **false positives 60%** and **test-infrastructure cost 40%**. Authored an internal white paper presented to Azure Core leadership.
- Reduced **throttling incidents 99%** for the platform's most critical customers and **92% across paying customers** by building fleet-wide telemetry pipelines, an early-warning dashboard and automated triage.
- Automated monthly business reviews from multi-day manual work into a governed **~5-minute reporting pipeline** consumed by leadership, reducing effort by **>95%** with independent reconciliation against Finance actuals.
- Shipped **15+ customer-facing recommendations** in Azure Advisor reaching **hundreds of thousands of accounts**; one moved **50% of targeted customers** off legacy storage.
- Delivered hands-on agentic AI workshops for **250+ engineers, data scientists and PMs per session** on skills, plugins and agents; wrote regular internal articles, mentored engineers and contributed to Azure Storage's cross-functional AI champion group.

Walmart Stores | Senior Data Scientist

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across **50 stores** with an end-to-end production **shelf-layout optimization** system, cutting planning cycles from weeks to hours.
- Built **product-association analytics** and **demand-transfer models** predicting demand shifts across products to inform assortment and replenishment decisions.
- Led **4 data scientists/analysts and 2 interns** to productionize the workflow for Merchandising Data Science; earned a Spot Award.

Integral Ad Science | Data Scientist Intern

Jun 2018 – Aug 2018

- Built a PySpark **text-classification pipeline** processing **~1M URLs/day** across 846 categories at 88% accuracy.

Mu Sigma | Data Scientist

Oct 2015 – Jul 2017

- Modeled **customer lifetime value** with probabilistic/regression models for Fortune-500 clients.

SKILLS

Analytics & ML: product analytics, metric design, statistical modeling, multi-armed bandits, optimization, anomaly detection, LLM-assisted analytics, revenue attribution, data validation, executive reporting

Engineering & Data: Python (pandas, scikit-learn, PyTorch, XGBoost), SQL, Spark/PySpark, FastAPI, Azure Data Explorer, Azure Data Factory, Azure OpenAI, MCP, Power BI, Azure DevOps, Git, data pipelines & governance

EDUCATION

University of Illinois at Chicago | MS, Business Analytics

2017 – 2018

SGSITS, Indore | BE, Computer Engineering

2011 – 2015